

"Global proteomic analysis of the lung in G2019S LRRK2 knockin mice under a specialized diet".

Dataset Description:

G2019S LRRK2 knockin mice and their wild-type littermates were maintained on a normal chow diet (23.7% kcal from fat) for the first 11 months of life. At that time, animals were randomly assigned to either a high-fat diet group (60% kcal from fat) or a control diet group (10% kcal from fat). Mice remained on their respective diets for an additional five months. At 16 months of age, mice were anesthetized with an isoflurane vaporizer, tissue were then collected and immediately snap frozen in liquid nitrogen. Frozen tissues were cryopulverized and homogenized. Protein were digested and dried down before resuspension for LC-MS/MS analysis. DIA data was processed in Spectronaut (version 18, Biognosys, Switzerland) using direct DIA. Data was searched against the Mus musculus proteome, including expected mutations. The manufacturer's default parameters were used. Each experimental group contained biological replicates (n = 4 per group).

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Dataset Name: lee-mouse-ms-p-lung-g2019s-hf-diet, v0.1

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ASAP Project: Are there bidirectional pathological interactions between senescence and LRRK2 models of PD

Project Description: Mutations in LRRK2 gene cause late-onset, autosomal dominant forms of familial PD that are clinically indistinguishable from sporadic PD. In addition, LRRK2 gene polymorphisms are associated with sporadic PD risk. The late disease-onset with these PD-linked genes suggest that interactions with age-related pathways are critical for manifesting or unmasking their pathogenic actions. Studies in experimental models further suggest that PD-linked LRRK2 mutations can induce abnormal mitochondrial function, lysosomal activity, autophagy, proteostasis and inflammatory processes, pathways that are dysregulated by cellular senescence. Our project focuses on determining whether: (1) PD-linked mutations in LRRK2 induce cellular senescence in the rodent brain, (2) increased senescence can exacerbate or unmask PD-like neurodegenerative phenotypes in LRRK2 mouse models, and (3) if removing senescent cells will provide neuroprotection in LRRK2 rodent models of PD.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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